

Project specifications

GOAL: Write and run a quantum algorithm with your **own hands**.

- Implementation of a quantum algorithm (e.g. Shor, Grover, Simon, quantum cryptography, error-correction, ...)
- Running the algorithm on two different backends: one noiseless and one with defined noise (Qiskit Aer, Quda-Q, real IBM quantum computer...)
- Outcomes: code (e.g. in the form of commented Jupyter notebook) and a protocol. Together, they must contain all information for understanding what was done, why it was done, and how exactly: The results have to be reproducible.
- Protocol:
 - As concise as possible. (Chat bots can be very talkative and I will be the only one reading the protocols, of course, its not forbidden to use them.)

DEADLINES:

- Deadline for assignment: 20/11/2025
- Deadline for submission of the protocol: 7/12/2025

Example of an assignment

Build a quantum circuit that can in a single run reveal a 4-bit hidden code s in the function $f(\{0,1\}^4) \rightarrow \{0,1\}$ whose form is $f(x) = s \cdot x \pmod{2} = s_0x_0 \oplus s_1x_1 \oplus s_2x_2 \oplus s_3x_3$. Implement 2 variants of this function with different s code using a quantum oracle constructed with CNOT and X quantum gates. Verify the designed quantum circuits on a simulator and then on a real quantum computer. The output of this work should therefore be two quantum circuits (two variants of the s code) and four measured results (each circuit needs to be verified on a simulator and also on a quantum computer). At the end of the work, it is necessary to evaluate the measured results.